A certified hot-spots family beyond the proven classes

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Abstract

Rauch's hot spots conjecture asks whether a second Neumann eigenfunction of the Laplacian attains its extrema only on the boundary. After fifty years the proofs cover special classes—domains with symmetry, lip domains, Euclidean triangles—while for convex quadrilaterals the question is open and current work addresses symmetric subcases. We prove the hot spots property for a single explicit domain outside every class with an analytic proof: the convex trapezoid with vertices $(0,0)$, $(1,0)$, $(17/20, 9/10)$, $(1/4, 9/10)$, which has no axis of symmetry and is not a lip domain. The proof is computer-assisted and fully certified: a deterministic chain of eight interval-arithmetic computations (re-runnable in under two minutes) that (i) localizes the second Neumann eigenvalue in [12.020976127, 12.022398359] and certifies its simplicity through Crouzeix–Raviart lower bounds with interval inertia counts; (ii) bounds the boundary defect of an exact 40-mode Fourier–Bessel trial by $1.2369 \cdot 10^{-5}$ using interval Taylor jets, giving $\|u - \hat{\varphi}\|_{L^2} \leq 6.5353 \cdot 10^{-5}$ for the true eigenfunction $\hat{\varphi}$; and (iii) excludes interior extrema on a three-part partition of the domain—a deep core by a solid-mean value argument with interior witnesses, a boundary collar by a swept centered-form evaluation with reflected pointwise bounds (zero residual cells), and four corner tips by certified Fourier–Bessel corner expansions, where the cold corner requires a Bessel ladder identity to evaluate a singular second derivative whose divergence has the helpful sign. We then extend the result from the point to an interval: the same property is certified for *every* c in $[0.845, 0.85]$, where $C = (c, 9/10)$, by seventeen chunks each carrying the full six-stage chain — to our knowledge the first certified positive-measure family of hot-spots domains outside the proven classes. Beyond the specific theorems, the paper documents a reusable instrument set for certified Neumann eigenfunction analysis on polygons and a ledger of failure modes of interval rigor that exact falsifiers caught during construction.

1 Introduction

In 1974 J. Rauch proposed what is now called the *hot spots conjecture* [1]: for a "nice" domain $\Omega \subset \mathbb{R}^2$, a second Neumann eigenfunction of the Laplacian attains its maximum and minimum on the boundary. The name records the physical picture: an insulated plate with generic initial heat distribution develops its hottest and coldest points on the edge, since the long-time temperature profile is dominated by the second Neumann mode.

*Independent researcher. Contact: carlos@carlostoledo.co. This work was produced with substantial AI assistance in a certified-verification setting: constructions and exposition were proposed with the help of large language models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for the certificate to be accepted. The certificates, not the models, are the evidence, and they are public and re-runnable. Machine-derived and not peer-reviewed.

The subsequent fifty years produced a landscape of partial results whose boundary is remarkably sharp. The conjecture is *false* in general: Burdzy and Werner constructed a planar counterexample with two holes [3], and Burdzy later one hole [4]; Kleefeld located numerical counterexample candidates with striking simplicity [10]; and de Dios Pont showed failure even for *convex* domains once the dimension is high enough [11], with quantitative refinements in [12]. It is *true* for domains with two axes of symmetry (Jerison–Nadirashvili [5]), for lip domains—domains between two Lipschitz graphs with constant ≤ 1—(Atar–Burdzy [6], following Bañuelos–Burdzy [2]), for classes of acute triangles (Siudeja [7]), and—in the landmark of the area—for all Euclidean triangles (Judge–Mondal [8]). For convex planar domains the conjecture is expected to hold, and Steinerberger proved the extrema live near the "tips" up to an inradius-scale neighborhood [9]; but beyond triangles no unconditional class is known. For convex *quadrilaterals*—the immediate frontier after [8]—the problem is open; current work treats symmetric subcases [13].

This paper proves the hot spots property for one explicit convex quadrilateral chosen to sit outside every proven class.

Theorem 1. *Let Ω be the trapezoid with vertices*

$$A = (0,0), \qquad B = (1,0), \qquad C = \left(\tfrac{17}{20}, \tfrac{9}{10}\right), \qquad D = \left(\tfrac{1}{4}, \tfrac{9}{10}\right),$$

and let φ be a second Neumann eigenfunction of $-\Delta$ on Ω (the second eigenvalue μ_1 is simple—part of the certificate—so φ is unique up to scale). Then φ attains its maximum and its minimum on $\partial\Omega$ only.

The domain is convex with side slopes 6 and 18/5; it has no axis of symmetry (so [5, 13] do not apply), both slanted sides are steeper than slope 1 (so it is not a lip domain in any orientation and [6] does not apply), and it is not a triangle. To our knowledge Theorem 1 makes Ω the first domain outside all analytically proven classes for which the hot spots property is a theorem. The result is a single specimen, not a class; its point is that the *certified-computation* route to such theorems now exists end to end, with the corner analysis—the part of [8] that consumed the most delicate analysis—carried out by interval arithmetic against exact corner expansions.

Method in one paragraph. Floating-point computation proposes; interval arithmetic certifies. A Crouzeix–Raviart eigenvalue computation with exact rational assembly and interval inertia counts brackets the low Neumann spectrum and proves μ_1 simple (§3). A method-of-particular-solutions trial u—a 40-term Fourier–Bessel sum over the four corners, an *exact* Helmholtz solution—is frozen, and its only defect, the Neumann boundary flux, is certified small by interval Taylor-jet quadrature (§4). A residual–gap argument in the spirit of Moler–Payne turns the defect into L^2, H^1 and eigenvalue enclosures for the true eigenpair (§5). A solid-mean lemma converts L^2 control of the error into pointwise control at interior points, which powers a value argument: certified suprema over a partition of the interior are beaten by certified values at two interior witness points (§6–7). What survives is a neighborhood of each corner, where φ equals its Fourier–Bessel corner expansion exactly; certified expansion coefficients kill three corners outright, and the cold corner falls to a no-critical-point argument plus a second-order Taylor bound off the Neumann edge whose input is a singular second derivative evaluated through a Bessel ladder identity (§8). Section 9 assembles the theorem; §11 records the methods ledger; Appendix A specifies the artifact and its trust base.

Relation to prior computational work. The trial-function technology descends from Fox–Henrici–Moler [17] and the eigenvector bounds of Moler–Payne [18], revived and stabilized by Betcke–Trefethen [19]; the certified lower bounds use the framework of Liu [14] in the form quoted by You–Xie–Liu [15] with the Crouzeix–Raviart element [16]. The two literature constants entering the certificate are Liu's framework theorem and the interpolation constant $0.1893\,h_K$; everything else is computed and interval-certified in the artifact. For the culture of computer-assisted proof by validated numerics see [20, 21]; unlike numerical-evidence studies of hot spots (e.g. [10]), every inequality below is a proved bound about the exact eigenfunction.

2 The specimen and the architecture of the proof

Write $0 = \mu_0 < \mu_1 \le \mu_2 \le \cdots$ for the Neumann eigenvalues of $-\Delta$ on Ω. The corner data of the specimen are exact:

vertex	coordinates	opening ω	(radians)	exponent $\nu = \pi/\omega$
A	$(0, 0)$	$\arctan(18/5)$	1.299849	2.41694
B	$(1, 0)$	$\arctan 6$	1.405648	2.23499
C	$(17/20, 9/10)$	$\pi - \arctan 6$	1.735945	1.80973
D	$(1/4, 9/10)$	$\pi - \arctan(18/5)$	1.841744	1.70587

All four openings are strictly less than π (convexity) and all exponents lie in $(1, 3)$: eigenfunctions are C^0 up to the corners, with first derivatives blowing up like $r^{\nu-1}$ only for $\nu < 1$ (never here) and second derivatives like $r^{\nu-2}$ at C and D (where $\nu < 2$)—the analytic fact that shapes both the choice of trial basis and the endgame at the cold corner.

The certified float landscape (a guide, not an input to any proof step): the boundary maximum of the normalized eigenfunction sits *at* vertex A with value ≈ 2.151 and the boundary minimum on the top edge near C at $\approx (0.8305, 0.9)$ with value ≈ -2.000.

Normalization. The certificate chain produces a frozen trial u with $\|u - c_1\varphi_1\|_{L^2} \le E$ for the L^2-normalized second eigenfunction φ_1 and some scalar c_1. All arguments concern $\hat{\varphi} := c_1\varphi_1 = u - e$; extrema locations are invariant under scaling, and the sign is fixed by $\hat{\varphi}(A) > 0$.

The partition. An interior extremum is excluded zone by zone (Figure 1):

- **Core** — depth ≥ 0.075 from $\partial\Omega$: a certified supremum over an adaptive cell cover is strictly beaten by a certified value at an interior witness (§6).
- **Collar** — depth < 0.075 and distance ≥ 0.11 from every vertex: the same value argument, cell by cell, with pointwise error bounds *reflected* across the nearby Neumann edge (§7).
- **Four tips** — distance ≤ 0.11 from a vertex: exact corner expansions with certified coefficients (§8).

Completeness of the partition is elementary given convexity (§9).

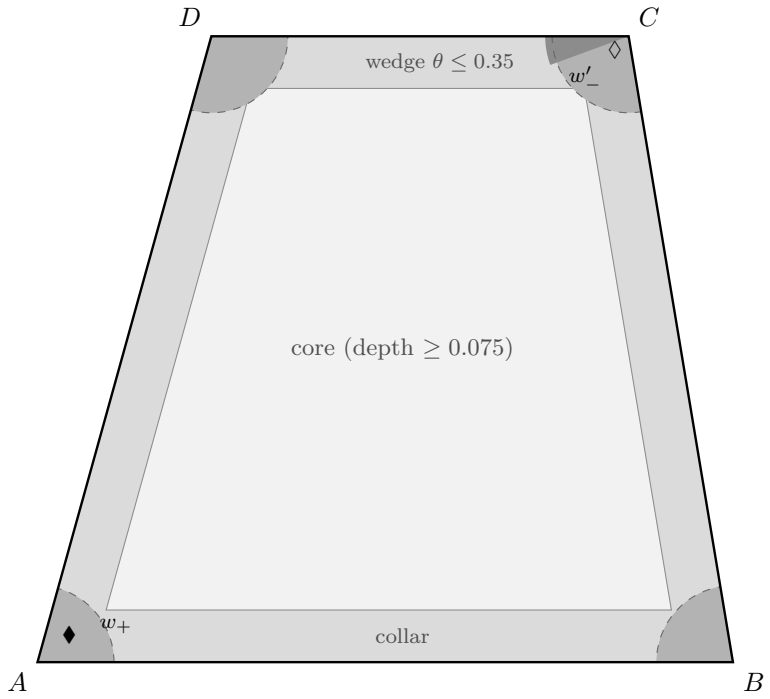


Figure 1: The zone map. Core: value argument against the witness w_+ (max side) and w'_- (min side), margins $5.4 \cdot 10^{-2}$ and $1.2 \cdot 10^{-2}$. Collar: 2,166 cells, reflected bounds, zero residual. Tips (radius 0.11, dashed): exact corner expansions — A dies radially, B and D by value, C by a no-critical-point sector plus the wedge, where $\hat{\varphi}_{nn} \ge 13.04$.

3 Certified spectrum: bracketing and simplicity

Upper bounds. A conforming Galerkin basis (products of cosines and shifted Legendre polynomials pulled back from the unit square, dimension 169) is assembled with *exact rational* integrals—the only non-polynomial integral family is expanded in a positive series with a certified tail—so the stiffness and mass matrices are known as rational intervals of width $\sim 10^{-16}$. Interval Rayleigh–Ritz (min–max over float-selected trial subspaces, evaluated in interval arithmetic) gives certified

$$\mu_1 \le 12.04181916, \qquad \mu_2 \le 14.12684697. \tag{1}$$

A regression on the rectangle $1 \times \frac{9}{10}$ reproduces $\mu_1 = \pi^2$ to all displayed digits.

Lower bounds. Nonconforming Crouzeix–Raviart [16] eigenvalues on a mesh with exact rational vertices bound the continuous spectrum from below through Liu's framework [14]: if $\|u - P_h u\|_{L^2} \le C_h \, |u - P_h u|_{1,h}$ then $\lambda_k \ge \lambda_{h,k}/(1 + C_h^2 \lambda_{h,k})$, with $C_h = 0.1893 \, h_{\max}$ from the per-triangle interpolation bound $\|u - \Pi_h u\|_{0,K} \le 0.1893 \, h_K \, |u - \Pi_h u|_{1,K}$ [15]. (In the Neumann configuration both spaces are taken mean-zero; the projection differs from Π_h by a constant, which only shrinks the L^2 norm.) The *discrete* eigenvalues are bounded by interval *inertia* counts: a diagonally pivoted right-looking interval elimination of $K - \sigma M$ combined with float symmetric scaling (both congruences, so Sylvester's law applies) counts eigenvalues below the

exact rational shift σ. The result, on a 456-dof mesh:

$$\mu_1 \in [11.892663,\ 12.04181916], \qquad \mu_2 \geq 13.955936, \tag{2}$$

and *exactly one* nonzero eigenvalue lies below 13.955936:

Proposition 1 (simplicity)**.** *μ_1 is a simple eigenvalue, and $\mu_2 - \mu_1 \geq 1.914$.*

This gap is the load-bearing input: it makes "the" second eigenfunction well defined and powers the eigenpair enclosure below.

4 The trial and its certified boundary defect

Why not Galerkin. A float diagnostic closed the naive route before rigor was built on it: for the smooth Galerkin basis the *strong* residual $\|-\Delta\tilde{u} - \lambda\tilde{u}\|_{L^2}$ *diverges* with basis size ($6.3 \to 9.7$ for $N = 8 \to 20$), driven by the corner singularities ($r^{\nu-2}$, $\nu \in (1.71, 2.42)$). Eigenvalue min–max is unaffected, but the eigenfunction certificate had to move to a basis with the corners built in.

The trial. The method of particular solutions [17, 19] with four corner Fourier–Bessel fans: at each vertex, modes $J_{k\nu}(\sqrt{\tilde{\lambda}}\, r)\cos(k\nu\theta)$, $k = 0, \ldots, 9$, in local polar coordinates (θ measured from one adjacent edge, ν the corner exponent). Each mode solves $-\Delta u = \tilde{\lambda} u$ *exactly* and has exact Neumann data on its two adjacent edges; the only defect of any linear combination is the Neumann flux on the far edges. The float scan (Betcke–Trefethen subspace angles, stabilized by rank-truncated SVD, column normalization, and an absolute noise floor; see §11) is deterministic and selects

$$\tilde{\lambda} = 12.021687243,$$

inside the certified bracket (2), with float defect $9.0 \cdot 10^{-6}$, stable from K=10 through K=18 per corner. The 40 coefficients at K=10 are frozen as exact doubles; from this point nothing about u is tuned.

Certified defect. On each edge, the flux of each far fan is a composition of J_ν, J_ν' and trigonometric factors of the local angle. Writing $s(\tau) = s_0 + \tau$ for arclength relative to a far corner (with constant transverse offset d, $r^2 = s^2 + d^2$, $\theta' = +d/r^2$), the flux $f(\tau)$ closes under differentiation via the Bessel ODE, so *order-2 interval Taylor jets* (f, f', f'') are computed without any hand-derived third-order chain rule. A certified cell rule

$$\textstyle\int_{\text{cell}} f^2 \ \leq\ h f(m)^2 + \frac{h^3}{12} f'(m)^2 + 2R\big(h|f(m)| + \frac{h^2}{4}|f'(m)|\big) + R^2 h, \qquad R = \frac{h^2}{8}\,\mathrm{mag}\, f''(\text{cell}),$$

summed over 700 cells per edge gives

$$\|\partial_\nu u\|_{L^2(\partial\Omega)} \leq D = 1.2369 \cdot 10^{-5}, \qquad \sup_{\text{edge}} |\partial_\nu u| \leq 1.4 \cdot 10^{-5} \text{ per edge}, \tag{3}$$

37% above the float value—the interval overhead is benign. Every jet order used is validated against an independent float evaluation ("bridges"), a guard installed after a sign bug in θ' inflated the certified defect 1000× *while all value bridges passed* (§11, item 5).

5 The eigenpair certificate

Green's identity turns the boundary defect into a weak residual: for $v \in H^1(\Omega)$, $\int_\Omega \nabla u \cdot \nabla v - \tilde\lambda uv = \int_{\partial\Omega} (\partial_\nu u)\, v$, so with the certified trace constant (star-shaped form, rational center $x_0 = (21/40,\ 9/20)$) $\|v\|_{L^2(\partial\Omega)} \le C_{\mathrm{tr}} \|v\|_{H^1}$, $C_{\mathrm{tr}} = 2.6427$, the expansion coefficients c_k of $u = \sum c_k \varphi_k$ obey

$$\sum_k c_k^2 \, \frac{(\mu_k - \tilde\lambda)^2}{1+\mu_k} \;\le\; \varepsilon^2, \qquad \varepsilon = C_{\mathrm{tr}}\, D = 3.2687 \cdot 10^{-5}. \tag{4}$$

Combining (4) with the spectrum localization (2) (exactly one nonzero eigenvalue below 13.955936) and the certified norm bound $\|u\|_{L^2} \ge c_1 \ge 0.166$ yields, with $F = \max\big(\tilde\lambda^{-2},\ (1+\mu_2^{\mathrm{lo}})/(\mu_2^{\mathrm{lo}} - \tilde\lambda)^2\big) = 3.9975$ and $G = \mu_2^{\mathrm{lo}}(1+\mu_2^{\mathrm{lo}})/(\mu_2^{\mathrm{lo}} - \tilde\lambda)^2 = 55.79$:

Proposition 2 (eigenpair enclosure)**.**

$$\begin{aligned} \mu_1 &\in [12.020976127,\ 12.022398359] && \textit{(width } 1.42 \cdot 10^{-3}\textit{, a } 105\times \textit{ tightening of CR)},\\ \|u - \hat\varphi\|_{L^2} &\le E = 6.5353 \cdot 10^{-5} = \sqrt{F}\,\varepsilon, && \textit{(relative } 3.9 \cdot 10^{-4}\textit{)},\\ |u - \hat\varphi|_{H^1} &\le \varepsilon\sqrt{G} = 2.4415 \cdot 10^{-4}. \end{aligned}$$

This is a self-contained residual–gap argument of Moler–Payne type [18]; the certified CR localization replaces the a priori spectral information assumed there.

6 Pointwise bounds and the core zone

L^2 control of $e = u - \hat\varphi$ becomes pointwise control through an explicit solid-mean identity, chosen over boundary-integral representations because it needs no Bessel-Y machinery: for $-\Delta w = f$ on $B_R(x_0) \subset \Omega$,

$$w(x_0) = \frac{1}{|B_R|}\int_{B_R} w + \int_{B_R} G_R f, \qquad G_R(r) = \frac{1}{2\pi}\Big(\ln\frac{R}{r} - \frac{R^2 - r^2}{2R^2}\Big), \qquad \|G_R\|_{L^2(B_R)} = R\sqrt{\frac{I_0}{2\pi}},$$

with $I_0 = \int_0^1 \big(-\ln t - \frac{1-t^2}{2}\big)^2 t\, dt = \frac{5}{48}$ *exactly* (derived by hand, reverified numerically in the artifact). Applied to e, which satisfies $-\Delta e = \mu_1 e + (\tilde\lambda - \mu_1)u$ with $|\tilde\lambda - \mu_1| \le 7.12 \cdot 10^{-4}$:

$$|e(x_0)| \;\le\; \frac{E}{\sqrt{\pi}\,R} + R\sqrt{\tfrac{I_0}{2\pi}}\Big(\mu_1^{\mathrm{up}} E + |\tilde\lambda - \mu_1|\,\|u\|_{L^2}\Big), \qquad B_R(x_0) \subset \Omega. \tag{5}$$

At depth 0.075 this certifies $|e| \le 5.5 \cdot 10^{-4}$; at the witness depths below, $\sim 10^{-3}$.

The value argument. No boundary values are ever needed. Two interior witnesses are placed just inside the boundary extrema: w_+ at depth 0.033 near A, and w'_- at depth 0.019 under the boundary minimum. Certified through (5):

$$\hat\varphi(w_+) \;\ge\; 2.126029, \qquad -\hat\varphi(w'_-) \;\ge\; 1.993811.$$

An adaptive branch-and-bound cover of the core (1,473 cells, centered form: cell value $\le$ center value + magnitude of cell gradient $\times$ half-diagonal) certifies

$$\sup_{\mathrm{core}} \hat\varphi \le 2.071924, \qquad \sup_{\mathrm{core}} (-\hat\varphi) \le 1.960684.$$

Lemma 1 (core zone)**.** *$\hat{\varphi}$ has no interior extremum at depth ≥ 0.075: an interior maximum would satisfy $\hat{\varphi}(x^*) \geq \hat{\varphi}(w_+) \geq 2.126029 > 2.071924$, a contradiction (margin $5.4 \cdot 10^{-2}$); symmetrically for the minimum (margin $1.2 \cdot 10^{-2}$).*

7 The collar

Cells at depth < 0.075 sit too close to $\partial\Omega$ for (5). The fix is reflection: across an open Neumann edge, the even reflection of $\hat{\varphi}$ extends it as an H^1 solution of the same eigenvalue equation, so solid-mean balls may cross the edge. The trial u is *not* exactly Neumann, so its reflection carries a single-layer correction with density $2\partial_\nu u$; its contribution to (5) is bounded by $\sup|\partial_\nu u| \cdot 3R/\pi$ using the certified per-edge sup in (3)—three orders below the budgets in play.

The sweep covers all 2,166 collar cells (depth < 0.075, distance ≥ 0.11 from every vertex) with the centered-form supremum plus reflected error bounds, on both sides ($\pm\hat{\varphi}$), with two refinement levels:

Lemma 2 (collar)**.** *Every collar cell has certified $\sup_{\text{cell}} \hat{\varphi} < 2.126029$ and $\sup_{\text{cell}}(-\hat{\varphi}) < 1.993811$: zero residual cells. No interior extremum lies in the collar.*

(The raw interval evaluation of the 40-term trial over a cell suffers catastrophic cancellation—widths ~ 1 at cell size 0.01; the centered form is what makes the sweep possible. See §11, item 4.)

8 The corner tips

In the sector of opening ω at each vertex, Neumann separation plus H^1 regularity (which excludes the singular Bessel-Y family) gives the *exact* representation

$$\hat{\varphi} \;=\; \sum_{k\geq 0} b_k \, J_{k\nu}(\sqrt{\mu_1}\, r)\cos(k\nu\theta), \qquad \nu = \pi/\omega, \tag{6}$$

convergent in the closed sector. The coefficients b_0, b_1, b_2 are certified by L^2 extraction over an annulus $r \in [0.03, 0.09]$: angular orthogonality makes the own-fan contribution of u exact, the far-fan part is integrated by second-order midpoint cells, and the contamination by e is controlled by E; a Parseval argument bounds the tail $\sum_{k\geq 3}$. Values, gradients and second derivatives of (6) are then interval-evaluated on each tip ($r \leq 0.11$) with certified geometric tails. Float guides: $b_0 \approx 2.151,\ 0.854,\ -2.000,\ -1.558$ at A, B, C, D—every tip has $|b_0| \geq 0.85$, so values separate cleanly from both extremal levels wherever the extremum is not adjacent.

Lemma 3 (tips A, B, D)**.** *On $r \leq 0.11$:*

B: $\hat{\varphi} \in [0.7566,\ 0.8965]$ — both sides dead by value.

D: $\hat{\varphi} \in [-1.6643,\ -1.3515]$ — both sides dead by value.

A: $\hat{\varphi} \in [2.0153,\ 2.2195]$ — min side dead by value; and $\partial_r \hat{\varphi} < 0$ throughout the sector (certified bound ≤ -0.0499; an inner disk factorization gives ≤ -10.0 near the vertex). An interior maximum at an interior point of the tip would be an interior critical point of $\hat{\varphi}$; there is none. Hence no interior extremum in tip A.

The cold corner C. The max side is dead by value ($\hat\varphi \le -1.8465$ on the tip). The min side is the delicate one: the boundary minimum lies on the top edge at distance ≈ 0.02 from the tip boundary, and the radial b_0/b_1 cancellation locus passes through the sector. The tip splits into two pieces, with θ measured from the top edge CD:

(i) The sector $\theta \in [0.35, \omega]$. The squared gradient of (6) is minorized by a quadratic form in $c = \cos(\nu\theta)$ (the "min-form"), interval-minimized in closed form over $c \in [-1, 1]$; this certifies $|\nabla\hat\varphi| > 0$ on $r \in [10^{-5}, 0.11]$, and an analytic leading-order argument covers $r < 10^{-5}$. No critical point, hence no interior extremum, in this piece.

(ii) The wedge $\theta \in [0, 0.35]$, $r \le 0.12$. Here the argument is transverse monotonicity from the Neumann edge. For a point at (signed) distance n from the top edge with foot a:

$$\hat\varphi(a, n) \;=\; \hat\varphi(a, 0) + \tfrac{n^2}{2}\,\hat\varphi_{nn}(a, \xi) \qquad \text{(no first-order term: Neumann)},$$

so a certified $\inf \hat\varphi_{nn} > 0$ over the wedge forces $\hat\varphi(\text{interior point}) > \hat\varphi(\text{foot}) \ge \min_{\partial\Omega} \hat\varphi$ — no interior wedge point can attain the minimum.

The input $\hat\varphi_{nn} = -\mu_1\hat\varphi - \hat\varphi_{tt}$ requires the tangential second derivative of (6), whose terms diverge like $r^{\nu-2}$ ($\nu = 1.80973$) with cancelling signs—exactly what interval arithmetic cannot see through: a naive polar-split Hessian returns widths that swallow the answer. The resolution is an identity that performs the cancellation analytically. Differentiating a Helmholtz corner mode twice along the fixed direction $\theta = 0$ and using the J recurrences:

$$\partial_t^2\Big[J_\nu(\sqrt{\mu}\,r)\cos(\nu\theta)\Big] = \frac{\mu}{4}\Big[J_{\nu+2}(\sqrt{\mu}\,r)\cos\big((\nu{+}2)\theta\big) + J_{\nu-2}(\sqrt{\mu}\,r)\cos\big((\nu{-}2)\theta\big) - 2J_\nu(\sqrt{\mu}\,r)\cos(\nu\theta)\Big], \tag{7}$$

one interval evaluation per mode, no subtraction of near-equal singular quantities—note the *negative* fractional order $\nu - 2 \approx -0.19$, which the special-function layer supports with closed-form falsifiers. The geometry cooperates: the singular $k{=}1$ term has $b_1 < 0$, so $\hat\varphi_{tt} \to -\infty$ and hence $\hat\varphi_{nn} \to +\infty$ as $r \to 0$—the corner singularity works *for* the bound.

Lemma 4 (tip C)**.** *On the tip at C: the maximum side is dead by value; on the wedge, $\hat\varphi_{nn} \ge 13.04$ (outer piece; inner piece ≥ 11.65), and no interior point of the sector piece is critical. No interior extremum lies in tip C.*

9 Assembly

Proof of Theorem 1. Partition completeness: every interior point has depth ≥ 0.075 (core), or depth < 0.075 and distance ≥ 0.11 from all vertices (collar), or distance < 0.11 from some vertex; in the last case convexity places it inside that vertex's sector, and the tip arguments run on the full sector $r \le 0.11$ (the wedge to 0.12).

Maximum. Suppose $x^* \in \Omega$ (interior) attains $\max_{\overline\Omega} \hat\varphi$. Then $\hat\varphi(x^*) \ge \hat\varphi(w_+) \ge 2.126029$. But x^* lies in the core (sup ≤ 2.071924, Lemma 1: contradiction), or a swept collar cell (certified sup < 2.126029, Lemma 2: contradiction), or tip B, C or D (values ≤ 0.8965, ≤ -1.8465 — on the relevant side — and ≤ -1.3515, Lemmas 3 and 4: contradiction), or tip A — where an interior maximum forces $\nabla\hat\varphi(x^*) = 0$, contradicting $\partial_r\hat\varphi < 0$ (Lemma 3).

Minimum. Symmetric, with w'_-: core ($1.960684 < 1.993811$), swept collar cells, tips A, B, D by value, and tip C by the no-critical-point sector plus the monotone wedge (Lemma 4).

Hence both extrema of $\hat\varphi$ — equivalently of any second Neumann eigenfunction, by simplicity (Proposition 1) and scale invariance — are attained on $\partial\Omega$ only. □

10 From a point to an interval: the band

A single certified domain invites the reply that it is a lucky specimen. This section removes that reply.

Theorem 2 (the band)**.** *For every $c \in [0.845,\, 0.85]$ let Ω_c be the trapezoid with vertices $A = (0,0)$, $B = (1,0)$, $C = (c, 9/10)$, $D = (1/4, 9/10)$. Then $\mu_1(c)$ is simple and the second Neumann eigenfunction of $-\Delta$ on Ω_c attains its maximum and its minimum on $\partial\Omega_c$ only.*

The interval is tiled by seventeen closed chunks — sixteen of width $3\cdot10^{-4}$ from 0.85 down to 0.8452, plus $[0.845, 0.8452]$ of width $2\cdot10^{-4}$ — each carrying the full six-stage chain of Sections 3–8. Adjacent chunks are closed and share endpoints, so their union is the closed interval with no gap. The specimen of Section 2 is $c = 17/20 = 0.85$ and is covered independently at the right endpoint.

Parametrisation. Per chunk $[c_{\mathrm{lo}}, c_{\mathrm{hi}}]$ put $c = c_{\mathrm{hi}} + \sigma H$ with $\sigma \in [-1, 0]$ and $H = c_{\mathrm{hi}} - c_{\mathrm{lo}}$. The trial family $u(x,\sigma) = \sum_i a_i(\sigma)\psi_i(x)$ has *fixed*, c-independent modes ψ_i anchored at the c_{hi} frames, with degree-three polynomial coefficients $a_i(\sigma)$; so u is an exact cubic in σ at every point and $-\Delta u = \tilde{\lambda}(\sigma)u$ holds exactly for every σ. Two facts made this work where a naive c-sweep does not. Raw multiple-precision-shooting coefficients are wild between nearby c — they vary by seven orders — so the family is solved once, point-normalised, rather than per-c. And frozen frames carry a branch cut along the frame ray, so the anchor is placed at the chunk's *right* end, keeping every cut ray outside every Ω_c in the chunk.

Certified values, worst over all seventeen chunks.

uniform spectrum	$\mu_1(c) \geq 11.85157$, $\mu_2(c) \geq 13.90774$
boundary defect	$D_{\sup} \leq 8.11 \cdot 10^{-5}$
eigenvalue window	$\mu_1(c) \in [12.01166,\, 12.04931]$
eigenpair error	$\|u_\sigma - c_1\varphi_{1,c}\|_{L^2} \leq 4.39 \cdot 10^{-4}$
zone margins	$\geq +2.47 \cdot 10^{-4}$ (max side), $\geq +6.41 \cdot 10^{-4}$ (min side)
collar survivors outside corner windows	0, every cell, every chunk
corner witnesses	$\hat{\varphi}(w_+) \geq 0.98692$, $-\hat{\varphi}(w'_-) \geq 0.92692$
tip C	$b_1 \leq -0.8791 < 0$ certified on every chunk

The uniform spectral pair gives simplicity for every c, and every eigenvalue window lies strictly inside (11.85157, 13.90774), so simplicity is never in question. The whole band cost about thirteen hours single-core, roughly forty-five minutes per chunk, with the defect stage dominating.

A resolution model, and its test. The binding resource is the number of σ-cells: the per-cell defect scales with cell width, so a coarse cell rides a multiple of the median defect and the zone stage fails first at a chunk's left end. Chunk $[0.8491, 0.8494]$ did exactly that, at worst margin $-1.1 \cdot 10^{-6}$. The model predicted that raising the cell count from 12 to 18 would buy about $+4\cdot10^{-3}$ of margin and sixteen or more chunks of runway. It then delivered: all fifteen remaining chunks certified on the first attempt, with zone margins between $+3.4 \cdot 10^{-3}$ and $+1.4 \cdot 10^{-2}$. We report this because a resolution model that predicts correctly is more reusable than the fix it recommended — and note that the two thinnest margins on the whole band belong to the two chunks built at the old resolution, not to any chunk the model governed.

What the band does not claim. It is an interval in *one* parameter of a four-parameter moduli space of convex quadrilaterals. It is not the quadrilateral conjecture, and it is not a census. The fence list of Section 12 applies unchanged.

11 Methods and lessons

The chain was built in one session against a standing test discipline: every certified computation ships with *exact falsifiers* — closed-form cases that must be contained, and independent float "bridges" for every quantity certified. Six failure modes were caught this way; we record them because each is generic to interval rigor, not specific to this problem.

1. **Intervals enter last.** In a rational-core pipeline, converting large intermediate quantities (10^{13}-sized Legendre monomial coefficients) to intervals *before* summation destroys exact cancellation: one Gram entry acquired width 3.94 (valid, but useless). Summing in exact rationals and converting once gives width 10^{-16}.

2. **Pivot interval factorizations.** Unpivoted interval LDL^{T} dies on a near-zero elimination pivot (an ordering artifact, row 211 of 456). Diagonal-pivoted right-looking elimination plus float symmetric scaling — both congruences, so inertia is preserved — is robust.

3. **Measure the float convergence curve before building rigor machinery.** The Galerkin strong residual diverges at the corners; a week of interval code on that route would have certified nothing. One diagnostic script killed it in advance and forced the MPS architecture.

4. **Centered forms, not raw cell intervals.** The raw interval value of a 40-term trial over a cell suffers multi-term cancellation blowup (widths ~ 1); the centered form (point value + cell gradient $\times$ half-diagonal) is what makes certified sup-covers feasible. Similarly, MPS needs rank-truncated SVD, column normalization, and an *absolute* noise floor — J_ν at large ν is float noise, and normalizing noise manufactures fake angles.

5. **Bridge every derivative order used.** A sign error in θ' left every *value* bridge passing (values came from the angle formula) while every τ-derivative used the wrong angular motion: term cancellation failed only in f' and the certified defect inflated $1000\times$. Value tests do not test jets.

6. **Clamp at domain guards.** Outward rounding pushes square-sums to $-5\cdot 10^{-324}$ and Γ's argument below its domain edge by one ulp; clamping (which only tightens) and extending by the functional equation are part of the rigor model.

12 Open problems

1. *The census.* Every instrument above is parameterized by the quadrilateral. The natural next step is a certified *census* over the moduli space of convex quadrilaterals — interval shape boxes, each box certified by the same chain with uniform constants — toward the statement that *convex quadrilaterals have no hot spots.* The obstructions are quantitative (near-degenerate gaps as symmetry is approached, where μ_1 ceases to be simple) rather than structural.

2. *Sharper μ_1.* The enclosure width $1.4 \cdot 10^{-3}$ is defect-limited; larger fans and finer jet quadrature should reach 10^{-8} at proportionate cost if the eigenvalue itself becomes the object of interest.

3. *Other specimens.* Domains further from every proven class — non-trapezoidal, near-degenerate, or with ν close to 1 at a cold corner — test which parts of the chain are generic. The cold-corner wedge argument used here needs only $b_1 \neq 0$ with the helpful sign; a specimen where it fails would need a genuinely new endgame.

A Reproducibility and trust base

The artifact is a directory of deterministic Node.js scripts (no dependencies beyond the bundled MIT-licensed interval/rational kernel); each prints an explicit `ok`/`FAIL` line per check and a final ALL-PASS verdict. The full chain reruns in ≈100 seconds single-core (measured on the build machine, 2026-09-01, all scripts ALL PASS):

script	role	certified output	secs
`test-ivspecial.js`	special-function layer	47 exact falsifiers	0.1
`run-p2a.js`	Galerkin uppers	$\mu_1 \leq 12.04181916$, $\mu_2 \leq 14.12684697$	7.1
`run-p2a2.js`	CR lowers + inertia	(2), simplicity	3.5
`run-p2b-mps.js`	float MPS trial (deterministic)	$\tilde{\lambda}$, frozen coefficients	15.2
`cert-defect.js`	interval Taylor-jet flux	(3)	4.6
`cert-assemble.js`	eigenpair enclosure	Proposition 2	4.4
`cert-pointwise.js`	solid-mean + core sweep	Lemma 1	6.8
`cert-collar.js`	reflected collar sweep	Lemma 2	7.3
`cert-corner.js`	corner expansions + tips	Lemmas 3, 4	49.9

Rigor model. Intervals are outward-rounded IEEE-754 doubles; exact quantities are BigInt rationals, converted to intervals as the *last* step of any derivation. The special-function layer (interval Γ by Spouge's formula with defensively widened error term; J_ν, J'_ν by power series with certified alternating tails, including negative fractional order) is tested against closed forms: Γ at integers and half-integers and its functional equation at fractional arguments; $J_{1/2}$, $J_{3/2}$ closed forms; the three-term recurrence at the specimen's actual ν values; derivative identities against central differences.

Literature inputs. Exactly two: Liu's framework theorem and the Crouzeix–Raviart interpolation constant $0.1893\, h_K$, both as stated in [15] (read against the source, not from memory).

Classical facts used. Neumann separation of variables in a sector together with H^1 regularity (excluding the Y_ν family) for (6); Green's identity on Ω; the spectral theorem; Courant's nodal bound. All are textbook-level and enter as stated hypotheses, not as computations.

Float landscape. All float numbers quoted (boundary extremum locations and values, coefficient guides) are labeled guides; no proof step consumes them.

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B An independent audit of the covering

An interval theorem is a union of chunk theorems, and such a union fails on *covering* far more readily than on arithmetic. Two covering claims carry the result and neither is visible inside any single certificate: the chunks must tile the parameter interval with shared endpoints, and inside each chunk the σ-cells must tile $[-1, 0]$ in every stage that reports per-cell numbers. A gap of 10^{-12} in either makes the interval statement false.

Both ladders were therefore re-derived from the stage records by a checker sharing no code with the program that produced them. It confirms that the 17 chunks tile $[0.845, 0.85]$ and that the 738 σ-cells tile $[-1, 0]$ in each of the zones, defect and eigenpair stages, and it re-derives every band-wide constant from per-cell data rather than reading a summary: $\mu_1 \geq 11.85157$, $\mu_2 \geq 13.90774$, thinnest zone margins $2.470e-4$ and $6.410e-4$, zero collar survivors outside the corner windows, and $\sup b_1 = -0.8791 < 0$ at corner C on every chunk. Corner C is identified by position rather than by the sign of b_1, since corner B also carries a negative b_1 and a sign-based scan could not detect C losing the property it is meant to verify.

Remark. The audit found 51 of the σ-cells to be *zero-width*, emitted where the refinement toward $\sigma = 0$ bottoms out and duplicates a shared endpoint. Such a cell certifies an empty set, so it cannot affect the covering, and the remaining cells still tile $[-1, 0]$ in every chunk and every stage. It is recorded rather than discarded because a certificate covering nothing is worth counting, and because the same rule that excludes these cells is what prevents a zero-width piece from papering over a genuine hole.